

Force and Pressure

Class 8 — Science

Learning Objectives

By the end of this chapter, students will be able to:

- Define force and identify different types of forces acting in everyday situations.
- Explain the relationship between force, area, and pressure using the standard formula.
- Describe how pressure varies in solids, liquids, and gases.
- Relate the concept of pressure to real-life phenomena such as walking on snow or using a knife.
- Distinguish between balanced and unbalanced forces and predict their effect on motion.

Prerequisites

Before starting this chapter, students should already be familiar with the concept of motion, the basic idea of mass as a measure of the amount of matter in an object, and simple unit conversions between grams and kilograms, and between centimetres and metres. A basic understanding of area calculation for squares and rectangles from Class 7 Mathematics is also assumed.

1. What is Force?

A force is simply a push or a pull that one object exerts on another. Force can change the speed of an object, change its direction of motion, or change its shape. When you push a door open, pull a drawer, or kick a football, you are applying a force. Forces are vector quantities, meaning they have both magnitude and direction, and they are measured in a unit called the newton (N), named after Sir Isaac Newton.

1.1 Types of Force

Forces can broadly be grouped into two categories based on whether physical contact is required between the objects involved.

- **Contact forces** — require direct physical contact, such as muscular force, frictional force, and the force applied while pushing a cart.
- **Non-contact forces** — act even when objects are not touching, such as gravitational force, magnetic force, and electrostatic force.

1.2 Balanced and Unbalanced Forces

When two or more forces acting on an object combine to produce zero net force, the forces are said to be balanced, and the object's state of motion does not change. When the combined effect of the forces is not zero, the forces are unbalanced, and the object accelerates in the direction of the resultant force. A tug-of-war in which neither team moves is a classic example of balanced forces.

2. Understanding Pressure

Pressure is defined as the force acting per unit area, applied in a direction perpendicular to the surface of an object. The relationship is expressed by the formula:

$$\text{Pressure (P)} = \text{Force (F)} / \text{Area (A)}$$

The SI unit of pressure is the pascal (Pa), where one pascal is equal to one newton of force acting on an area of one square metre ($1 \text{ Pa} = 1 \text{ N/m}^2$). From this formula, an important relationship follows: for a constant force, pressure increases as the area decreases, and pressure decreases as the area increases. This explains why a sharp knife cuts more easily than a blunt one — the smaller contact area concentrates the same force into much greater pressure.

2.1 Pressure in Fluids

Liquids and gases are collectively referred to as fluids, and both exert pressure on the walls of their container and on any object immersed in them. Liquid pressure increases with depth, which is why dams are built thicker at the base than at the top. Gas pressure, such as atmospheric pressure, arises from the continuous collision of gas molecules with a surface, and it decreases with increasing altitude, which is why the air becomes 'thinner' on tall mountains.

Table 1: Everyday Objects and Their Pressure Behaviour

Object	Approx. Contact Area	Effect on Pressure
Sharp needle	Very small	Very high pressure for small force
Wide ski board	Large	Low pressure, prevents sinking in snow
School bag with wide straps	Larger than thin straps	Reduces pressure on shoulders
Camel's broad feet	Large	Distributes weight, low pressure on sand

3. Real-Life Applications

- Wide foundations of buildings distribute the building's weight over a larger area, reducing pressure on the ground.
- Railway tracks are laid on wooden or concrete sleepers to spread the pressure exerted by trains over a larger area.
- The nib of a fountain pen is kept thin so that the ink can be pushed out under sufficient pressure.
- Tractors and army tanks are fitted with broad tyres/tracks to reduce pressure on soft or muddy ground.

4. Common Misconceptions

Misconception: A heavier object always exerts more pressure than a lighter one.

Clarification: Pressure depends on both force and area — a heavier object with a very large contact area may exert less pressure than a lighter object concentrated on a small point, such as a stiletto heel versus a flat shoe of the same weight.

Misconception: Force and pressure are the same physical quantity.

Clarification: Force is a push or pull measured in newtons, while pressure is force distributed over an area, measured in pascals. Two objects can experience the same force but very different pressures depending on their contact area.

Important Note

Note: Always express area in square metres (m^2) before calculating pressure in pascals. A common numerical error in examinations is forgetting to convert centimetres to metres before applying the pressure formula.

Summary

Force is a push or pull that can change an object's speed, direction, or shape, and is measured in newtons. Pressure is the force acting per unit area, measured in pascals, and is governed by the relationship $P = F/A$. Reducing the area over which a given force acts increases pressure, while spreading a force over a larger area reduces pressure. These principles explain a wide range of everyday phenomena, from the design of building foundations to the shape of skis and knives.

Reference: National Council of Educational Research and Training — Science Curriculum Resources, <https://ncert.nic.in/textbook.php>